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Chapter 4

Polynomial Functions

4.1 Linear function

Definition 4.1.1. A function f is said to be linear if it is of the form $y = ax + b$, where a and b are real numbers.

4.1.2 Linear equations and solutions

Definition 4.1.3. An equation is a statement that two algebraic expressions are equal.

They include

1. $3x - 5 = 7$

2. $\frac{x}{3} + \frac{3x}{4} = 2$.

To solve an equation in x means to find all values of x for which the equation is TRUE. Such values are called **solutions**.

4.1.4 Linear equations in one variable

Definition 4.1.5. A linear equation in one variable is an equation that can be written in the standard form

$$ax + b = 0,$$

where a and b are real numbers with $a \neq 0$.

A linear equation in x has exactly one solution. For example, from $ax + b = 0$, we get that $ax = -b$, so that $x = -\frac{b}{a}$.

Example 4.1.6. Solve each of the following equations

1. $3x - 6 = 0$

2. $5 + 5x = 15$

3. $10 - x = 11$.

4.2 Equations involving fractional expressions

To solve an equation involving fractional expressions you can multiply every term in the equation by the least common denominator (LCD) of the terms.

Example 4.2.1. Solve each of the following fractional equations

1. $\frac{x}{3} + \frac{3x}{4} = 2$.

2. $\frac{4x}{5} - \frac{x}{2} = 9$.

An equation with a single fraction on both sides can be cleared of denominators by **cross-multiplying**, which is equivalent to multiplying each side of the equation by the least common denominator and then simplify.

Example 4.2.2. Cross-multiply to solve each of the following fractional equations

1. $\frac{3y-2}{2y+1} = \frac{6y-9}{4y+3}$.

2. $\frac{3x-6}{x+10} = \frac{3}{4}$

4.2.3 Graphs of linear functions

Since any function can be represented by ordered pairs, (x, y) , a graphical representation of the function is always possible. The graph of an

equation in the two variables x and y is the set of all points (x, y) whose coordinates satisfy the equation.

To graph a linear function, substitute various values of x into the equation and solve for y to produce some of the ordered pairs that satisfy the equation.

Example 4.2.4. Draw the graph of each of the following by plotting the points.

1. $y = 2x - 1$

2. $y = -x + 3$

3. $y = 4$

4. $x = 3$.

4.3 Quadratic functions

Definition 4.3.1. A function f , in x , is said to be quadratic if it is of the form $f(x) = ax^2 + bx + c$, where a, b and c are real numbers with $a \neq 0$.

4.3.2 Quadratic equations

Definition 4.3.3. A quadratic equation in the variable x is defined as the equation of the form $ax^2 + bx + c = 0$, where a, b and c are constants with $a \neq 0$.

Such equations can be solved using the following methods:

1. Factorization

2. Quadratic formula. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

3. Completing the square.

For the purpose of this study, we will only look at method (3), unless you want us to revisit the other two.

- Example 4.3.4.** 1. Use factorization method to solve $3x^2 - 5x + 2 = 0$.
2. By using the quadratic formula, solve $2x^2 + 7x + 4 = 0$.

To solve a quadratic equation ($ax^2 + bx + c = 0$) by completing the square method, the following steps will be required.

1. If a does not equal 1, divide each side by a so that the coefficient of x^2 is 1.
2. Rewrite the equation with the constant term on the right side.
3. Complete the square by adding the square of one-half of the new coefficient of x to both sides.
4. Write the left side as a square and simplify the right side.
5. Finally, simplify both sides so that you can take square root of both sides.

The following example will help us understand the above steps.

Example 4.3.5. Using completing the square method, solve

1. $3x^2 - 5x + 2 = 0$
2. $2x^2 + 7x + 4 = 0$.

By using the quadratic formula, we observe that, we obtain different solutions depending on the value of the radicand $b^2 - 4ac$. $b^2 - 4ac$ is called the **discriminant** of the quadratic equation. The discriminant is used to determine the nature of solutions/roots of a quadratic equation as follows:

1. If $b^2 - 4ac > 0$, then the quadratic equation has two distinct real roots.
2. If $b^2 - 4ac < 0$, then the quadratic equation has no real roots (has complex roots).

3. If $b^2 - 4ac = 0$, then the quadratic equation has one real root.

Example 4.3.6. Determine the nature of roots for the following equations:

1. $4x^2 - 7x - 1 = 0$

2. $4x^2 + 12x + 9 = 0$.

3. $5x^2 + 2x + 1 = 0$.

4. $9x^2 - 16 = 0$.

5. $2x^2 + 2x - 2 = 4x$.

4.4 Graph of a quadratic function

The following steps will be needed to be able to sketch the graphs of quadratic functions.

1. Decide on the shape.

(a) When $a > 0$, the curve will be a $\cup$ shape.

(b) When $a < 0$, the curve will be a $\cap$ shape.

2. Work out the points where the curve crosses the x – and y –axes.

(a) Put $y = 0$ to find the point(s) where it crosses the x –axis.

(b) Put $x = 0$ to find the point where it crosses the y –axis.

The following examples will help us understand the steps above.

Example 4.4.1. Sketch the graph of each of the following.

1. $f(x) = x^2 - 5x + 4$.

2. $f(x) = -2x^2 - 7x + 4$.

By the process of completing the square, all quadratic functions, $f(x) = ax^2 + bx + c$, $a \neq 0$, may be transposed into what will be called the turning point form;

$$y = a(x - h)^2 + k, \text{ where } (h, k) \text{ is the turning point.}$$

We determine the x - and y - intercepts as before. i.e. by equating $y = 0$ and $x = 0$ respectively.

Example 4.4.2. Sketch the graph of each of the following:

1. $f(x) = x^2 - 5x + 4$.

2. $f(x) = -2x^2 - 7x + 4$.

3. Let $p(x) = -\frac{x^2}{10} + 50x - 750$ be the profit function in dollars that a company earns as a function of x , number of products of a given type that are sold, and is valid for values x greater than or equal to 0, and less than or equal to 500. Find the maximum loss and maximum profit.

4.5 Polynomial functions

In this section, we will study polynomial functions and rational functions.

Definition 4.5.1. A polynomial function is a function defined by

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_1 x + a_0.$$

For instance, $f(x) = x^3 + 4x^2 - x + 1$ is a polynomial function, and is called a third-degree or cubic polynomial function because the largest power is 3.

Definition 4.5.2. A rational function is a function defined by the quotient of two polynomials.

For example,

$$f(x) = \frac{7x}{2x^2 + 5}$$

is a rational function.

If p is a polynomial function, then the values of x for which $p(x)$ is equal to 0, is called the zeros (solutions) of p . For instance, -1 is a zero of $p(x) = 2x^3 - x + 1$ because $p(-1) = 0$.

Much of this section concerns finding the zeros of polynomial functions. Sometimes the zeros of a polynomial function are determined by dividing one polynomial by the other.

4.5.3 Division of polynomials

Consider the polynomial $p(x) = x^3 - 2x^2 + 4x - 3$. Divide this polynomial by $x - 3$. Using long division as in arithmetic, we get the following result

$$\frac{x^3 - 2x^2 + 4x - 3}{x - 3} = (x^2 + x + 7) + \frac{18}{x - 3},$$

and the steps are as follows.

1. Divide x^3 by x to give x^2 .
2. Multiply $x - 3$ by x^2 .
3. Subtract to get x^2 and bring down the next term, $4x$.
4. Divide x^2 by x to give x .
5. Multiply $x - 3$ by x .
6. Subtract to get $7x$ and bring down the next term, -3 .
7. Divide $7x$ by x to give 7 .
8. Multiply $x - 3$ by 7 .
9. Subtract; this gives the remainder 18.

Thus

$$\frac{x^3 - 2x^2 + 4x - 3}{x - 3} = (x^2 + x + 7) + \frac{18}{x - 3}.$$

Hence

$$f(x) = (x^2 + x + 7)(x - 3) + 18.$$

From the example above, we see that, when we divide a polynomial $f(x)$ by $(x - a)$, we get the quotient Q and the remainder R , so that

$$f(x) = Q \times (x - a) + R,$$

where $R = f(a)$. In general, if the polynomial $f(x)$ is divided by the linear expression $px + q$, then

$$f(x) = Q \times (px + q) + R,$$

where $R = f(\frac{-q}{p})$. This is called the **remainder theorem** for a polynomial $f(x)$.

Note that the theorem only applies to polynomials with linear divisors.

Example 4.5.4. What is the remainder when the polynomial $f(x) = x^3 - x^2 + 3x - 2$ is divided by;

1. $x - 1$
2. $x + 2$
3. $2x - 1$?

Example 4.5.5. 1. The polynomial $f(x) = x^3 + ax^2 - 3x + 4$ is divided by $x - 2$ and the remainder is 14. What is the value of a ?

2. Let $f(x) = x^3 + ax^2 + bx - 3$. When $f(x)$ is divided by $x - 1$ and $x + 1$, the remainder is 1 and -9 respectively. Find the value of a and of b .

4.5.6 The factor theorem

If $(x - a)$ is a factor of $f(x)$, then the remainder when $f(x)$ is divided by $(x - a)$ is zero. That is; $f(a) = 0$. Similarly, if $px + q$ is a factor of $f(x)$, then $f\left(\frac{-q}{p}\right) = 0$. This is called the factor theorem for a polynomial $f(x)$.

Example 4.5.7. Determine whether $x + 1$ is a factor of;

1. $f(x) = x^2 + 2x + 1$.
2. $f(x) = x^3 - 6x^2 - x + 6$.
3. $f(x) = x^3 - 2x^2 - x + 2$.

Example 4.5.8. Given that $p(x) = x^3 + kx^2 + x + 6$, find the value of k if $x + 1$ is a factor of p . Hence, find the other factors.

Example 4.5.9. Factorize $x^3 - 6x^2 - x + 6$.

Solution

As $f(x)$ is of degree 3, it will have atmost three linear factors of the general form $px + a$, $qx + b$ and $rx + c$, so that

$$\begin{aligned}x^3 - 6x^2 - x + 6 &= (px + a)(qx + b)(rx + c) \\ &= pqr x^3 + \dots + abc\end{aligned}$$

Since the coefficient of x^3 is 1 and the last term is +6, we observe that $pqr = 1$ and $abc = +6$. So the possible factors come from the factors of +6. The first factor has to be found by trial which in this case is $x - 1$.

We could continue like this to get the other factors but the best method is to use long division, considered earlier and show that $x^3 - 6x^2 - x + 6 = (x - 1)(x^2 - 5x - 6) = (x - 1)(x + 1)(x - 6)$. We could also use synthetic division stated below;

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0,$$

where $a_i \neq 0$ for all i .

Given a polynomial

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

we can divide it by a linear factor $x - a$, where ' a ' is a real number, using the following steps.

1. Write the a on the left side of a vertical bar and the coefficients in decreasing degree as follows;

$$x = a \mid a_n \quad a_{n-1} \quad \cdots \quad a_1 \quad a_0$$

2. Pass the coefficient of the highest degree below the horizontal line as below

$$\begin{array}{rcccccc} & a \mid a_n & a_{n-1} & \cdots & a_1 & a_0 \\ \hline & a_n & & & & \end{array}$$

3. Multiply the passed coefficient by a and add to the next coefficient

$$\begin{array}{rcccccc} & a \mid a_n & a_{n-1} & \cdots & a_1 & a_0 \\ & a_n & aa_n + a_{n-1} & & & \end{array}$$

4. Continue step 3 until all coefficients are covered.
5. The numbers below $a_n, a_{n-1}, \dots, a_1$ give the coefficients, in decreasing order, of the quotient whose degree is 1 less than that of the polynomial. The number below a_0 is the remainder.

Example 4.5.10. Solve each of the following:

1. $x^3 - 6x^2 + 8x = 0$

$$2. x^3 - 2x^2 - x + 2 = 0$$

Example 4.5.11. Sketch each of the following

$$1. f(x) = x^3 - 6x^2 + 8x$$

$$2. f(x) = x^3 - 2x^2 - x + 2$$

4.6 Rational functions; Partial fractions

Definition 4.6.1. A rational function $R(x)$ is the quotient of two polynomials $N(x)$, $D(x)$ such that $R(x) = \frac{N(x)}{D(x)}$ provided that $D(x) \neq 0$.

$R(x)$ is called proper if the degree of $N(x)$ is less than the degree of $D(x)$, or equivalently, if $N(x)$ is smaller than $D(x)$. If $N(x)$ is greater than $D(x)$, then $R(x)$ is called an improper rational function.

4.6.2 Partial fractions

In this section, we will learn how to split a rational function into its partial fractions. First we consider the following example.

Example 4.6.3. Express $\frac{1}{x+2} + \frac{2}{x-3}$ as a single fraction.

This gives

$$\frac{1}{x+2} + \frac{2}{x-3} = \frac{3x+1}{(x+2)(x-3)} = \frac{3x+1}{x^2-x-6}.$$

$\frac{1}{x+2}$ and $\frac{2}{x-3}$ are called partial fractions of $\frac{3x+1}{x^2-x-6}$, and the ability to represent a complicated algebraic fraction in terms of its partial fractions is the purpose of this section.

Rules of partial fractions

1. The numerator must be of lower degree than the denominator. That is, it must be a proper rational function. If it is not, then we first divide out.

2. Factorize the denominator into its prime factors. These determine the “shape” of the partial fractions.

The following list summarizes the forms of all the possible partial fractions.

$$\frac{A}{x+a} + \frac{B}{(x+a)^2} + \frac{C}{(x+a)^3}$$

Denominator containing	Expression	Form of partial fraction
Linear factors	$\frac{f(x)}{(x+a)(x-b)(x+c)}$	$\frac{A}{x+a} + \frac{B}{x-b} + \frac{C}{x+c}$
Repeated linear factors	$\frac{f(x)}{(x+a)^2(x+b)}$	$\frac{A}{x+a} + \frac{B}{(x+a)^2} + \frac{C}{x+b}$
Quadratic factors	$\frac{f(x)}{(ax^2+bx+c)(x+d)}$	$\frac{Ax+B}{(ax^2+bx+c)} + \frac{C}{x+d}$

The following examples will help us understand the above discussion.

Example 4.6.4. Split the following into partial fractions.

1. $\frac{x}{(x+2)(x+3)}.$
2. $\frac{11-3x}{x^2+2x-3}.$
3. $\frac{2x^2-9x-35}{(x+1)(x-2)(x+3)}.$
4. $\frac{x^3-2x^2-4x-4}{x^2+x-2}.$
5. $\frac{2x+3}{(x-2)^3}.$
6. $\frac{3+6x+4x^2-2x^3}{x^2(x^2+3)}.$

Example 4.6.5. Express the following as a sum of its partial fractions.
partial fractions:

1. $\frac{11-3x}{x^2+2x-3}.$

2. $\frac{11-3x}{x^2+2x-3}$.
3. $\frac{2x^2-9x-35}{(x+1)(x-2)(x+3)}$.
4. $\frac{x^3-2x^2-4x-4}{x^2+x-2}$.
5. $\frac{2x}{(x-2)^2(x+2)}$.
6. $\frac{x^2+2}{(x+2)^2(x+3)}$.

Example 4.6.6. Decompose each of the following into its partial fractions.

1. $\frac{1}{x(x^2+5)}$.
2. $\frac{3+6x+4x^2-2x^3}{x^2(x^2+3)}$.

Example 4.6.7. Split into partial fractions:

1. $\frac{x^4+1}{x^3+9x}$.
2. $\frac{6x^3+x^2+5x-1}{x^3+x}$.

4.7 Inequalities

4.7.1 introduction

Simple inequalities are used to order real numbers. The inequality symbols $<$, $\leq$, $>$ and $\geq$ are used to compare two real numbers and to denote subsets of real numbers. For example, the simple inequality $x \geq 3$ denotes all real numbers x that are greater than or equal to 3.

In this section, we will expand our work with inequalities to include more involved statements such as

$$5x - 7 > 3x + 9 \text{ and } -3 \leq 6x - 1 < 3.$$

As is the case with equations, we solve an inequality in the variable x by finding all values of x for which the inequality is true. Such values are

solutions and are said to satisfy the inequality. The set of all real numbers that are solutions of an inequality form a solution set. For example, the solution set of $x + 3 > 4$ is

$$S = \{x : x > 1, x \in \mathbb{R}\}.$$

4.7.2 Solving linear inequalities

The simplest type of inequality to solve is a linear inequality in x . For example $2x + 3 > 4$ is a linear inequality in x .

As we solve the following examples, remember that when you multiply and divide by a negative number, you must reverse the inequality symbol.

Example 4.7.3. Solve each of the following linear inequalities.

1. $5x - 7 > 3x + 9$

2. $3x < 2x + 1$

3. $1 - \frac{3x}{2} \geq x - 4.$

4. $-3 \leq 6x - 1 < 3.$

4.7.4 Quadratic inequalities

Definition 4.7.5. A quadratic inequality is any inequality that can be put in one of the forms

1. $ax^2 + bx + c < 0$

2. $ax^2 + bx + c > 0$

3. $ax^2 + bx + c \leq 0$

4. $ax^2 + bx + c \geq 0$

where a, b and c real numbers and $a \neq 0$.

Procedure for solving Quadratic inequalities

- Express the given inequality in the standard form.
- Solve the equation $ax^2 + bx + c = 0$. The real solutions are the boundary points.
- Locate these boundary points on a number line, thereby dividing the number line into test intervals.
- Choose one representative number within each test interval. If substituting that value into the original inequality produces a true statement, then all the real numbers in that test interval belong to the solution set. Otherwise, no real numbers in the test interval belong to the solution set.
- Write the solution set; the intervals that produce a true statement.

Example 4.7.6. Solve and graph the solution on a number line.

1. $2x^2 - 3x \geq 2$.

2. $5x^2 - 9x > 5$.

Example 4.7.7. Sketch the graph of $-x^2 + 2x \geq -5$. Find the values of x for which $-x^2 + 2x + 5 \geq 0$.

4.7.8 Polynomial inequalities

Procedure for solving polynomial (cubic) inequalities

- Express the given inequality in the standard form.
- Solve the equation $ax^3 + bx^2 + cx + d = 0$. The real solutions are the boundary points.
- Locate these boundary points on a number line, thereby dividing the number line into test intervals.

- Choose one representative number within each test interval. If substituting that value into the original inequality produces a true statement, then all the real numbers in that test interval belong to the solution set. Otherwise, no real numbers in the test interval belong to the solution set.
- Write the solution set; the intervals that produce a true statement.

Example 4.7.9. Solve and graph the solution on a number line.

1. $x^3 + x^2 \geq 8x + 12$.

2. $x^3 + 8 \leq 5x^2 - 2x$.

Example 4.7.10. Sketch each of the following, and hence state the set(s) of value of x for which $f(x) \geq 0$.

1. $f(x) = x^3 - 6x^2 + 8x$

2. $f(x) = x^3 - 2x^2 - x + 2$

4.7.11 Rational inequalities

Definition 4.7.12. Asymptote

An asymptote is a line that a graph gets closer and closer to, but never touches or crosses it.

To solve a rational inequality:

- Get all the terms on the left hand side.
- Combine all the terms into one single rational expression.
- Factor the ration expression so you know the exact location of the zeroes and vertical asymptotes, and thus where the y -values change sign.
- State the solution as the interval(s) that have the desired y -values as demanded by the inequality.

Example 4.7.13. Solve the inequalities:

1. $\frac{1}{x+5} < \frac{x-2}{x-7}$.

2. $\frac{x+1}{x+3} \leq 2$.

3. $\frac{x+5}{x^2-7x+12} \leq 0$.

4.7.14 Graph of rational functions

Definition 4.7.15. Asymptotes A nonvertical line with equation $y = mx + c$ is called an asymptote of the graph of $y = f(x)$ if the difference $f(x) - (mx + c)$ tends to 0 as x takes on arbitrarily large values of x ; $x \rightarrow \pm\infty$.

If $m = 0$ then $y = b$ is called a horizontal asymptote. Otherwise $y = mx + b$ is called a slant asymptote.

A vertical line $x = a$ is called a vertical asymptote if $|f(x)|$ takes arbitrarily large values as $x \rightarrow a$ from the right or left.

Finding the horizontal/slant asymptotes.

1. Degree of numerator less than that of the denominator; H.A is $y = 0$. Here, no S.A.
2. Degree of numerator equal to that of the denominator; H.A is $y = \text{ratio of the degree of the leading coefficients}$. Here, no S.A.
3. Degree of numerator greater than that of the denominator;
 - (a) By 1; H.A non. To get slant A. use long division.
 - (b) By more than 1; No H.A or S.A

Example 4.7.16. Find the asymptotes for each of the following:

1. $f(x) = \frac{-2x^2}{2x^3+4x^2}$. Hint; H.A $y = 0$, V.A $x = -2$ and a hole at $x = 0$, and no slant/oblique asymptote.

2. $g(x) = \frac{6x^2+x+12}{3x^2-5x-2}$. Hint; H.A $y = \frac{6}{3} = 2$, V.A $x = -\frac{1}{3}, 2$.

3. $h(x) = \frac{2x^3 - 5x^2 + x - 7}{x^2 + x - 3}$. Hint; only S.A/O.A along $y = 2x - 7$ after long division.

4. $f(x) = \frac{x^4 - 5x^2 + 4}{x^2 + 2x + 1}$. Hint V.A at $x = -1$ and a hole at $x = -1$.

In order to successfully sketch the graph of a rational function, the following analysis will be useful.

1. Find the domain of the rational function. Use the excluded values from the domain to form the vertical asymptotes of the graph.
2. Find the x and y intercepts.
3. Test for symmetry;
 - (a) y -axis symmetry when the function is even.
 - (b) origin symmetry occurs when the function is odd.
4. Reduce the function if possible. A hole in the graph will result if the function reduces.
5. Find the H.A or S.A asymptotes. Determine if the graph intersects them
6. Determine if the graph is above or below the x -axis using the x -intercepts to divide the x -axis into intervals.
7. Sketch the graph using the above information.
 - (a) First, graph the asymptotes.
 - (b) Second, graph the intercepts and the points.
 - (c) Third, verify the end behavior.

Example 4.7.17. Analyze the graph of each of the following functions.

1. $y = \frac{1}{x}$.

$$2. y = \frac{1}{x^2}.$$

$$3. y = \frac{1}{x+3}$$

$$4. f(x) = \frac{x+2}{x^2-9}.$$

4.8 Modulus function

The modulus of a function, which is also called the absolute value of a function gives the magnitude and absolute value of a number irrespective of the number being positive or negative. It always gives a non-negative value of any number or variable.

Definition 4.8.1. The modulus function f , in the variable x , is the function defined by

$$|f(x)| := \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

4.8.2 Graph of the function $y = |f(x)|$

To sketch the graph of the modulus function, $y = |f(x)|$, the following simple steps will be useful.

1. Sketch the function $y = f(x)$; ignoring the modulus. For the part of the curve below the x -axis (the negative values of y ,) reflect in the x -axis.

Example 4.8.3. Sketch the graph of each of the following;

$$1. y = |x|.$$

$$2. y = |3x - 2|.$$

$$3. y = x^2 - 3x - 10.$$

Example 4.8.4. 1. (a) On the same diagram, sketch the graphs of $y = |5x - 1|$ and $y = |2x|$.

(b) Solve the equation $|5x - 1| = |2x|$.

2. (a) On the same diagram, sketch the graphs of $y = |x^2 - 2x|$ and $y = \frac{1}{4} - 2x$.

(b) Solve the equation $|x^2 - 2x| = \frac{1}{4} - 2x$

4.8.5 Other transformations

Some basic transformations useful in sketching of graphs of different functions are;

1. $f(x + a)$ is a horizontal translation of $-a$.
2. $f(x) + a$ is a vertical translation of $+a$.
3. $f(ax)$ is a horizontal stretch of scale factor $\frac{1}{a}$.
4. $af(x)$ is a vertical stretch of scale factor a .

Example 4.8.6. 1. Sketch the graph of $y = (x - 2)^2 + 3$.

2. Sketch the graph of $\frac{2}{x+5}$.